



Splunk® Enterprise Search Tutorial 7.3.5

Use a subsearch

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Use a subsearch

In this section you will learn how to correlate events by using subsearches.

A subsearch is a search that is used to narrow down the set of events that you search on. The result of the subsearch is then used as an argument to the primary, or outer, search. Subsearches are enclosed in square brackets within a main search and are evaluated first.

Let's find the single most frequent shopper on the Buttercup Games online store, and what that shopper has purchased.

The following examples show why a subsearch is useful. Example 1 shows how to find the most frequent shopper without a subsearch. Example 2 shows how to find the most frequent shopper with a subsearch.

Example 1: Search without a subsearch

You want to find the single most frequent shopper on the Buttercup Games online store and what that shopper has purchased. Use the `top` command to return the most frequent shopper.

1. Start a new search.
2. Change the time range to **All time**.
3. To find the shopper who accessed the online shop the most, use this search.

```
sourcetype=access_* status=200 action=purchase | top limit=1 clientip
```

The `limit=1` argument specifies to return 1 value. The `clientip` argument specifies the field to return.



The screenshot shows the 'New Search' interface in Splunk. The search bar contains the query: `sourcetype=access_* status=200 action=purchase | top limit=1 clientip`. Below the search bar, it indicates '5,224 events (before 4/18/19 10:24:29.000 AM)' and 'No Event Sampling'. The results are displayed in a table with the following data:

clientip	count	percent
87.194.216.51	134	2.565084

This search returns one `clientip` value, 87.194.216.51, which you will use to identify the VIP shopper. The search also returns a count and a percent. These are the default fields that are returned with the `top` command.

4. You now need to run another search to determine how many different products the VIP shopper has purchased. Use the `stats` command to count the purchases by this VIP customer.

```
sourcetype=access_* status=200 action=purchase clientip=87.194.216.51 | stats count, distinct_count(productId), values(productId) by clientip
```

This search uses several statistical functions with the `stats` command. An alias for the `distinct_count()` function is `dc()`.

The screenshot shows the 'New Search' interface in Splunk. The search bar contains the query: `sourcetype=access_* status=200 action=purchase clientip=87.194.216.51 | stats count, distinct_count(productId), values(productId) by clientip`. The results are displayed in the 'Statistics (1)' tab. The table shows one row for the client IP 87.194.216.51, with a count of 134, 14 distinct products, and a list of those products.

clientip	count	distinct_count(productId)	values(productId)
87.194.216.51	134	14	BS-AG-G09 CU-PG-G06 DB-SG-G01 DC-SG-G02 FI-AG-G08 FS-SG-G03 MB-AG-G07 MB-AG-T01 PZ-SG-G05 SC-HG-G10 WC-SH-A01 WC-SH-A02 WC-SH-G04 WC-SH-T02

This search uses the `count()` function to return the total count of the purchases for the VIP shopper. The `dc()` function is the `distinct_count` function. Use this function to count the number of different, or unique, products that the shopper bought. The `values` function is used to display the distinct product IDs as a multivalue field.

The drawback to this approach is that you have to run two searches each time you want to build this table. The top purchaser is not likely to be the same person in every time range.

Example 2: Search with a subsearch

Let's start with our first requirement, to identify the single most frequent shopper on the Buttercup Games online store.

1. Copy and paste the following search into the Search bar and run the search. Make sure the time range is **All time**.

```
sourcetype=access_* status=200 action=purchase | top limit=1 clientip | table clientip
```

This search returns the clientip for the most frequent shopper, `clientip=87.194.216.51`. This search is almost identical to the search in Example 1 Step 1. The difference is the last piped command, `| table clientip`, which displays the clientip information in a table. Because you specified only the `clientip` field with the `table` command, that is the only field returned. The count and percent fields that the `top` command generated are discarded from the output.

To find what this shopper has purchased, you run a search on the same data. You provide the result of the most frequent shopper search as one of the criteria for the purchases search.

The most frequent shopper search becomes the **subsearch** for the purchases search. The purchases search is referred to as the **outer** or primary search. Because you are searching the same data, the beginning of the outer search is identical to the beginning of the subsearch.

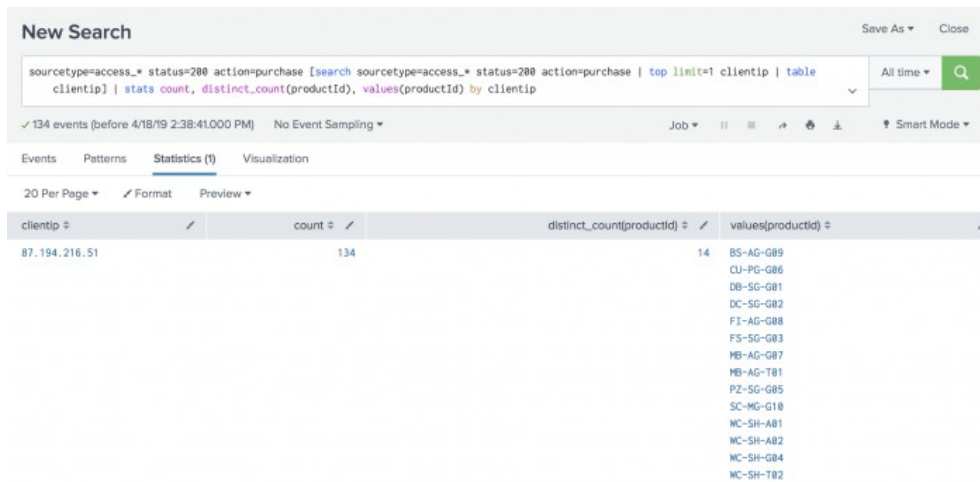
A subsearch is enclosed in square brackets `[ ]` and processed first when the search criteria are parsed.

2. Copy and paste the following search into the Search bar and run the search.

```
sourcetype=access_* status=200 action=purchase [search sourcetype=access_* status=200  
action=purchase | top limit=1 clientip | table clientip] | stats count, distinct_count(productId),
```

```
values(productId) by clientip
```

Because the `top` command returns the **count** and **percent** fields, the `table` command is used to keep only the `clientip` value.



The screenshot shows the Splunk Search interface. The search bar contains the query: `sourcetype=access_* status=200 action=purchase [search sourcetype=access_* status=200 action=purchase | top limit=1 clientip | table clientip] | stats count, distinct_count(productId), values(productId) by clientip`. The results table has four columns: `clientip`, `count`, `distinct_count(productId)`, and `values(productId)`. The first row shows the IP address `87.194.216.51` with a count of `134` and a list of 14 product IDs.

clientip	count	distinct_count(productId)	values(productId)
87.194.216.51	134	14	BS-AG-G89 CU-PG-G86 DB-SG-G81 DC-SG-G82 FI-AG-G88 FS-SG-G83 MB-AG-G87 MB-AG-T81 PZ-SG-G85 SC-MG-G10 MC-SH-A81 MC-SH-A82 MC-SH-G84 MC-SH-T82

These results should match the result of the two searches in Example 1, if you run it on the same time range. If you change the time range, you might see different results because the top purchasing customer will be different.

The performance of this subsearch depends on how many distinct IP addresses match `status=200 AND action=purchase`. If there are thousands of distinct IP addresses, the `top` command has to keep track of all of those addresses before the top 1 is returned, impacting performance. By default, subsearches return a maximum of 10,000 results and have a maximum runtime of 60 seconds. In large production environments, it is possible that the subsearch in this example will timeout before it completes. The best option is to rewrite the query to limit the number of events that the subsearch must process. Alternatively, you can increase the maximum results and maximum runtime parameters.

Make the search syntax easier to read

Subsearches and long complex searches can be difficult to read. You can apply auto-formatting to the search syntax to make the the search syntax easier to read in the Search bar. Use the following keyboard shortcut to apply auto-formatting to a search.

- ◆ On Linux or Windows use **Ctrl + **.
- ◆ On Mac OSX use **Command + **. You can also use **Command + Shift + F**, which works well with many non-English keyboards.



The screenshot shows the Splunk Search interface with the search query: `sourcetype=access_* status=200 action=purchase [ search sourcetype=access_* status=200 action=purchase | top limit=1 clientip | table clientip ] | stats count, distinct_count(productId), values(productId) by clientip`. The query is formatted with color-coding and indentation to improve readability.

clientip	count	distinct_count(productId)	values(productId)
87.194.216.51	134	14	BS-AG-G89 CU-PG-G86 DB-SG-G81 DC-SG-G82 FI-AG-G88 FS-SG-G83 MB-AG-G87 MB-AG-T81 PZ-SG-G85 SC-MG-G10 MC-SH-A81 MC-SH-A82 MC-SH-G84 MC-SH-T82

Make the search results easier to understand

You can make the information more understandable by renaming the columns.

Column	Rename
count	Total Purchased
dc(productId)	Total Products
values(productId)	Product IDs
clientip	VIP Customer

You rename columns by using the AS operator on the fields in your search. If the rename that you want to use contains a space, you must enclose the rename in quotation marks.

3. To rename the fields, copy and paste the following search into the Search bar and run the search.

```
sourcetype=access_* status=200 action=purchase [search sourcetype=access_* status=200  
action=purchase | top limit=1 clientip | table clientip] | stats count AS "Total Purchased",  
distinct_count(productId) AS "Total Products", values(productId) AS "Product IDs" by clientip |  
rename clientip AS "VIP Customer"
```

The screenshot shows the Splunk Search interface. At the top, the search bar contains the query: `sourcetype=access_* status=200 action=purchase [search sourcetype=access_* status=200 action=purchase | top limit=1 clientip | table clientip] | stats count AS "Total Purchased", distinct_count(productId) AS "Total Products", values(productId) AS "Product IDs" by clientip | rename clientip AS "VIP Customer"`. Red boxes highlight the `AS "Total Purchased"`, `AS "Total Products"`, `AS "Product IDs"`, and `AS "VIP Customer"` parts of the query. Below the search bar, the results are displayed in a table view. The table has four columns: `VIP Customer`, `Total Purchased`, `Total Products`, and `Product IDs`. Red boxes highlight these column headers. The `VIP Customer` column shows the IP address `87.194.216.51`. The `Total Purchased` column shows the value `134`. The `Total Products` column shows the value `14`. The `Product IDs` column lists 14 product IDs: `BS-AG-G09`, `CU-PG-G06`, `DB-SG-G01`, `DC-SG-G02`, `FI-AG-G08`, `FS-SG-G03`, `MB-AG-G07`, `MB-AG-T01`, `PZ-SG-G05`, `SC-MG-G10`, `WC-SH-A01`, `WC-SH-A02`, `WC-SH-G04`, and `WC-SH-T02`.

4. Experiment with this search. What happens when you run the search over different time periods? What if you wanted to find the top product sold and how many people bought it?

Next step

This completes Part 4 of the Search Tutorial.

You have learned how to use fields, the Splunk search language, and subsearches to search your data. Continue to Part 5: Enriching events with lookups.

See also

About subsearches in the *Search Manual*

The top command in the *Search Reference*

The stats command in the *Search Reference*